AS91946: Interpret and Apply Mathematical and Statistical Information

NCEA Level 1 Statistics — Year 11

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1. Welcome

This guide combines the learning for both *AS 1.1* (91944: Explore data using a statistical enquiry process) and *AS 1.3* (91946: Interpret and apply statistical information). By highlighting where these two standards overlap, this resource eliminates unnecessary duplication and shows you how core statistical thinking connects the data you collect with the data you evaluate in everyday life.

1.1. One toolkit, two assessments

You are not learning two separate subjects. The same toolkit appears in both courses:

- **AS1.1 (Internal)** - statistical investigation in context
- **AS1.3 (External / 91946)** - interpret and apply statistical information in context

In the internal assessment, you use that toolkit to plan an investigation, collect data, analyse it, and write a conclusion. In the external assessment, you use the same ideas to read graphs, interpret tables, question claims, and justify your reasoning under exam conditions.

1.2. Where the standards overlap

Area	AS1.1 Internal	AS1.3 External (91946)	Overlap?
Statistical enquiry cycle (PP-DAC)	Core assessed process	Used for interpretation and critique	**Both**
Designing and carrying out an investigation	Strong focus	Not the primary assessment focus	**Mostly 1.1**
Reading graphs/tables in context	Required	Required	**Both**
Numerical summaries (mean, median, IQR, spread)	Used for evidence in investigation	Directly assessed in interpretation questions	**Both**

Area	AS1.1 Internal	AS1.3 External (91946)	Overlap?
Probability, two-way tables, expected value	Limited or indirect	Directly assessed	**Mostly 1.3**
Justification and statistical reasoning	Required in conclusion and evaluation	Required in written responses	**Both**
Exam-style interpretation under time pressure	Not the main format	Core format	**Mostly 1.3**

1.3. How this book is organised

This draft keeps the shared statistical foundation together first, then moves probability and critical reading into a second part.

Part	Main focus	Why it comes here
Part 1	Statistical questions, data types, displays, comparison, and practice	These chapters support both standards immediately.
Part 2	Probability and critical reading of statistical information	These ideas matter for AS1.3, but they can sit after the core foundation.

1.4. Assessment tags used in this book

- [**1.1**] mainly assessed in the internal
- [**1.3**] mainly assessed in the external
- [**Both**] directly useful for both standards

1.5. Suggested learning pathway

Follow this sequence from start to finish:

Order	Learning focus	Where in this book	Tested in
1	Combined overview of AS1.1 and AS1.3 expectations	Chapter 1 - Combined approach	[**Both**]

Order	Learning focus	Where in this book	Tested in
2	Statistical questions, context language, core terms	Chapter 2 - What is Statistics?	[**Both**]
3	Statistical enquiry cycle and investigation flow	Chapter 3 - PPDAC	[**Both**]
4	Variables and data types, choosing suitable displays	Chapter 4 - Data types	[**Both**]
5	Graph interpretation and comparison language	Chapter 5 - Graphs	[**Both**]
6	Centre and spread for comparison and conclusions	Chapter 6 - Measures	[**Both**]
7	Applied tasks and mixed practice with response scaffolds	Chapter 7 - Practice	[**Both**]
8	Probability tools and table-based reasoning	Part 2 - Probability	[**1.3**]
9	Sampling quality, bias, inference, critique	Part 2 - Critical reading	[**Both**]
10	Extension reading and source material	Resource Library	[**Both**]

1.6. How to use this pathway for dual preparation

1. Build investigation capability first (Chapters 1-4) for strong **AS1.1** evidence.
2. Use the practice chapter to consolidate comparison and communication.
3. Add the Part 2 extension chapters for explicit **AS1.3** preparation.
4. In every written response, include context, comparison, and justification language.

1.7. What this standard is about

In AS1.3 (91946), you will learn to:

- **Read and interpret** statistical displays, tables, and graphs
- **Apply** mathematical and statistical ideas to real-world problems
- **Communicate** findings clearly using statistical language
- **Evaluate** statistical claims made in media and everyday life

1.8. How to use this book

Each chapter builds on the previous one. You will find:

- Clear explanations of key concepts with New Zealand examples
- Interactive R code blocks in the HTML version showing how statistics is done with real data
- Practice questions at Achievement, Merit, and Excellence levels
- Glossary boxes highlighting important vocabulary

In the PDF version, the working code is hidden so the focus stays on the mathematics and interpretation. In the HTML version, the code can still be opened for teaching, demonstration, or extension.

At a Glance: AS1.3 (External)

Grade	What you need to do
Achieved	Interpret and apply statistical information
Merit	Interpret and apply statistical information with justification
Excellence	Interpret and apply statistical information with statistical insight

Internal/External connection

If you can explain findings from your own investigation clearly and in context (AS1.1), you are building the same reasoning needed for interpretation and justification in AS1.3.

1.9. Tools we use

All data visualisations and calculations in this book are produced using **R** and rendered with **Quarto**. You do not need to be able to write R code to succeed in this standard. The code is there so your teacher can demonstrate the analysis live, and so students who want extension material can inspect how the graphs and summaries were built.

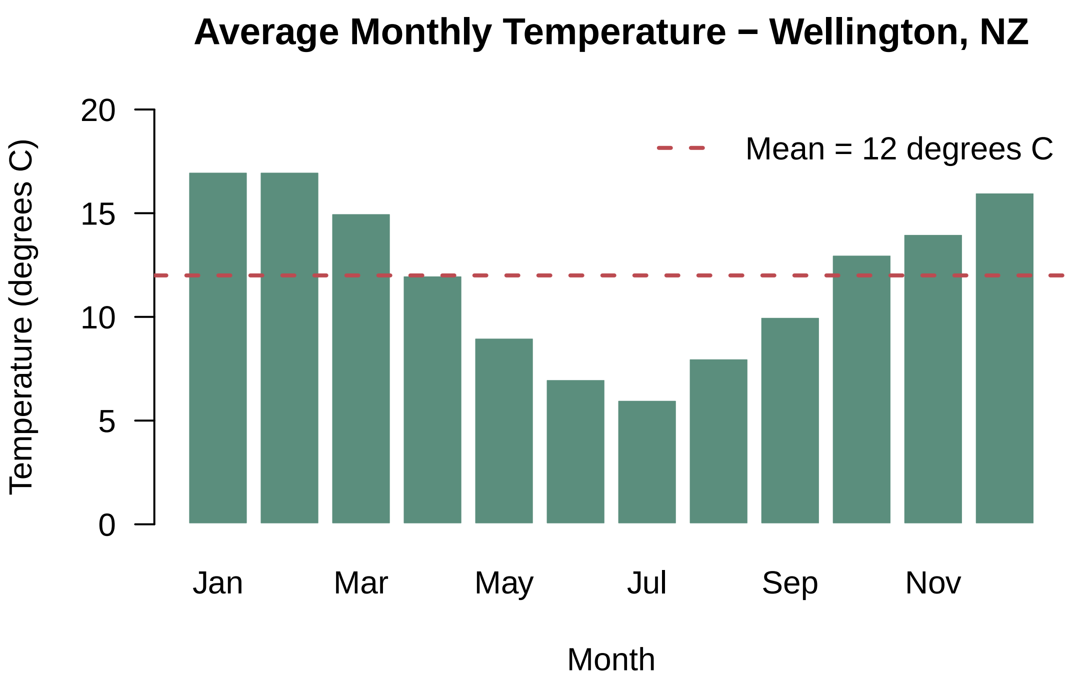

Figure 1.1.: An example of the kind of visualisations we will create in this book.

1.10. If statistics feels new

Start with the core chapters and keep a small set of support resources nearby:

- Khan Academy for quick refreshers on averages, graphs, and probability
- CensusAtSchool New Zealand for local student data examples
- Stats NZ for authentic graphs, tables, and media-style summaries
- The Resource Library at the end of this book for curriculum and standard documents

1.11. Getting started

Head to Chapter 3 to begin the combined 1.1 + 1.3 pathway.

Part I.

Part 1: Core Statistical Thinking

2. Combined AS1.1 and AS1.3 Approach

This chapter explains how this book combines the shared teaching for **AS91944/1.1 (Internal)**—*Explore data using a statistical enquiry process*—and **AS91946/1.3 (External)**—*Interpret and apply mathematical and statistical information.*

The aim is to make overlap explicit, reduce duplication, and show how one set of statistical ideas supports both assessments.

2.1. One toolkit, two assessments

You are not learning two separate subjects. The same toolkit appears in both courses:

- **AS91944/1.1 (Internal)**

Grade	What you need to do
Achieved	Explore data using a statistical enquiry process.
Merit	Explore data using a statistical enquiry process, with justification.
Excellence	Explore data using a statistical enquiry process, with statistical insight.

- **AS91946/1.3 (External)**

Grade	What you need to do
Achieved	Interpret and apply statistical information.
Merit	Interpret and apply statistical information with justification.
Excellence	Interpret and apply statistical information with statistical insight.

ℹ Note

In AS1.1, you use that toolkit to plan an investigation, collect data, analyse it, and write a conclusion. In AS1.3, you use the same ideas to read graphs, interpret tables, question claims, and justify reasoning under exam conditions.

2.2. Where the standards overlap

Area	AS1.1 Internal	AS1.3 External (91946)	Overlap?
Statistical enquiry cycle (PP-DAC)	Core assessed process	Used for interpretation and critique	**Both**

Area	AS1.1 Internal	AS1.3 External (91946)	Overlap?
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Justification and statistical reasoning	Required in conclusion and evaluation	Required in written responses	**Both**
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2.3. Links to the rest of this text

Use this chapter as the map for what each section contributes to dual preparation:

Section	Main learning link	Standards link
Chapter 3	General foundations: statistical language, context, population/sample/variable	**Both**
Chapter 4	Statistical enquiry cycle and investigation logic	**Both** (especially AS1.1 process)
Chapter 5	Variable classification and choosing appropriate displays	**Both**

Section	Main learning link	Standards link
Chapter 6	Interpreting and comparing displays in context	**Both**
Chapter 8	Centre, spread, and comparison statements	**Both**
Part 2 chapters	Probability and critical reading under assessment-style conditions	**Mostly 1.3**

2.4. Getting started

Continue to Chapter 3 for the general introduction to statistics, then move to Chapter 4 for the dedicated enquiry-cycle chapter.

2.4.1. Suggested learning pathway

1. Build statistical language and context first (Chapter 3).
2. Learn the enquiry process (Chapter 4) before writing conclusions.
3. Strengthen representation by understanding data types (Chapter 5) and choosing appropriate graphs (Chapter 6).
4. Use measures of centre and spread to compare distributions and justify findings (Chapter 8).
5. Consolidate all core skills with mixed practice questions (Chapter 9).
6. Extend to probability tools (Chapter 10) and critical reading of statistical claims (Chapter 11) for explicit AS1.3 preparation.

> **i Internal and external connection**
>
> If you can explain findings from your own investigation clearly and in context (AS1.1), you are building the same reasoning needed for interpretation and justification in AS1.3.

3. What is Statistics?

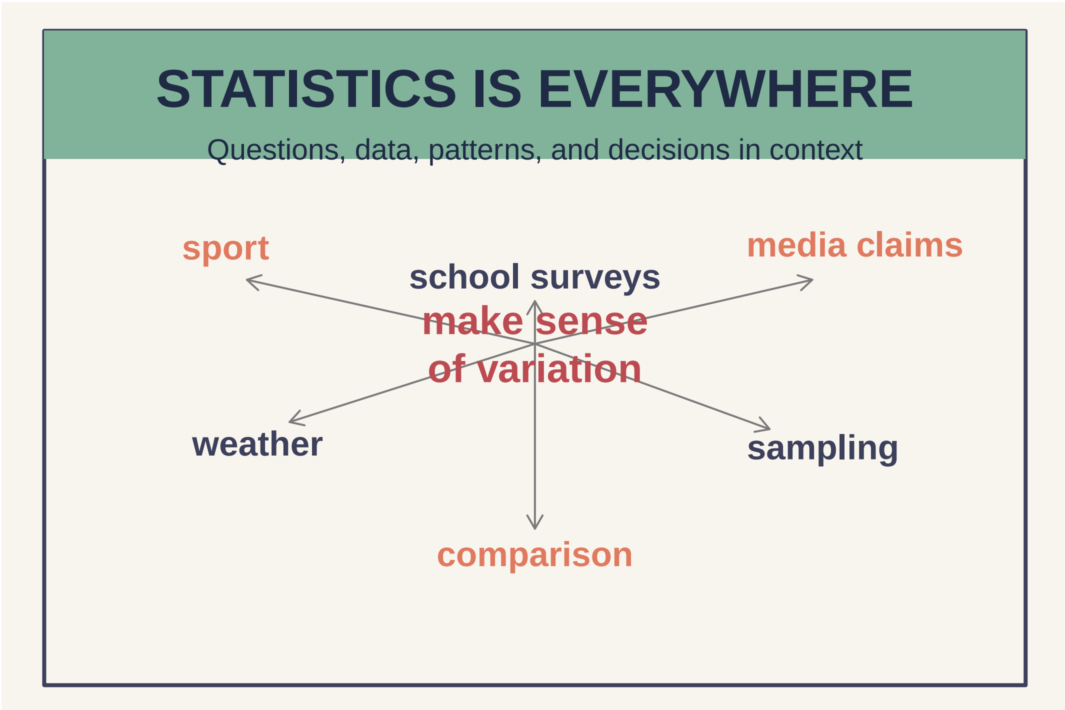

Figure 3.1.: Statistics connects questions, data, and decisions in real contexts.

3.1. Why statistics matters

Every day we are surrounded by data. News headlines cite poll results, sports commentators compare player averages, health campaigns report risk percentages, and social media shows us trending topics. **Statistics** is the science that helps us make sense of all this information.

At school, statistics helps you ask better questions, decide what evidence is relevant, and explain what the data does or does not support. Outside school, it helps you read media claims critically instead of accepting every graph or headline at face value.

> ℹ Key definition
>
> **Statistics** is the practice of collecting, organising, summarising, analysing, and interpreting data in order to draw conclusions and make decisions.

3.2. Statistical questions and variability

Not all questions are statistical questions. A statistical question expects **variability** in the answers.

Tip

Statistical question: "How tall are the students in Year 11?"
(Heights will vary - this is statistical.)
Not a statistical question: "How tall is Ms Smith?"
(Only one answer - not statistical.)

3.2.1. Practice identifying statistical questions

Which of the following are statistical questions?

1. How many pages are in Harry Potter and the Philosopher's Stone?
2. How many hours per week do Year 11 students spend on homework?
3. What is the population of New Zealand?
4. What are the most popular music genres among teenagers in Auckland?
5. What colour is the school uniform?
6. How long does it take students in our class to get to school?
7. Which lunch option is chosen most often in the canteen each Friday?
8. How many goals did the first XI football team score this season?
9. What is the average screen time for Year 11 students on a school night?
10. How many desks are in Room 12?

Answer key

1. Not statistical - there is one fixed answer.
2. Statistical - different students spend different amounts of time.
3. Not statistical - there is one (changing but definite) answer.
4. Statistical - different teenagers have different preferences.
5. Not statistical - the uniform colour is fixed.
6. Statistical - travel times will vary between students.
7. Statistical - different students make different choices.
8. Not statistical - the season total is one fixed value.
9. Statistical - students will report different screen times.
10. Not statistical - there is one fixed count for that room.

3.2.2. A template for writing your own statistical question

Use this sentence frame when you are stuck:

> What is the distribution of [**variable**] for [**population**] in [**context**]?

For example:

- What is the distribution of weekly study time for Year 11 students at our school?
- How does travel time to school differ for students who take the bus compared with students who walk?

3.3. Population, sample, and variable

These three ideas appear constantly in both the internal and the external.

Idea	What it means	School-based example
Population	The full group you want to know about	All Year 11 students at the school
Sample	The smaller group you actually collect data from	The 30 Year 11 students surveyed in form time
Variable	What you measure or record	Height, travel time, or favourite sport

⚠ Common trap

A large sample is not automatically a good sample. If the sample is biased, the conclusion can still be misleading.

3.4. Why context matters

In AS91946, **context** is central. You will always work with data that relates to real situations. When interpreting statistics, you must:

- **Name the variable** being measured
- **State the units** (e.g. centimetres, dollars, minutes)
- **Refer to the population** the data came from
- **Relate findings back** to the real-world situation

The strongest answers do more than quote a number. They explain what that number means for the people, objects, or events being studied.

⚠ Common mistake

Saying "the mean is 24" is not enough. Say
"The mean age of survey respondents was 24 years."

3.4.1. Population vs sample in context

Suppose a school wants to know how much sleep Year 11 students get on a normal school night.

- The **population** is all Year 11 students at the school.
- A **sample** might be one statistics class or 40 students chosen at random.
- The **variable** is hours of sleep on a school night.

If the sample only comes from one extension class, the results might not represent the whole year level. That is why context and sampling always matter when you interpret a summary or graph.

3.5. Glossary

Term	Meaning
Data	Values collected by observation or measurement
Variable	A characteristic that can take different values
Population	The entire group of interest

Term	Meaning
Sample	A subset of the population selected for study
Statistic	A numerical summary calculated from a sample
Parameter	A numerical summary of a whole population
Context	The real-world situation that gives meaning to the data

3.6. Exit ticket

1. Give two examples of statistical questions you could investigate at your school.
2. A student says "the average score is 65". What is missing from this statement?
3. What is the difference between a population and a sample?
4. Write one sentence that uses a summary statistic in full context.
5. Explain why a sample could be large but still poor quality.

Exit ticket answers

1. Any valid examples should expect variation, such as study time, travel time, or sleep.
2. The variable, the units, and the group being described are missing.
3. The population is the whole group of interest; the sample is the smaller group actually measured.
4. Example: "The median travel time for Year 11 students in our survey was 18 minutes."
5. A large sample can still be biased if it leaves out important groups or is collected unfairly.

4. The Statistical Enquiry Cycle (PPDAC)

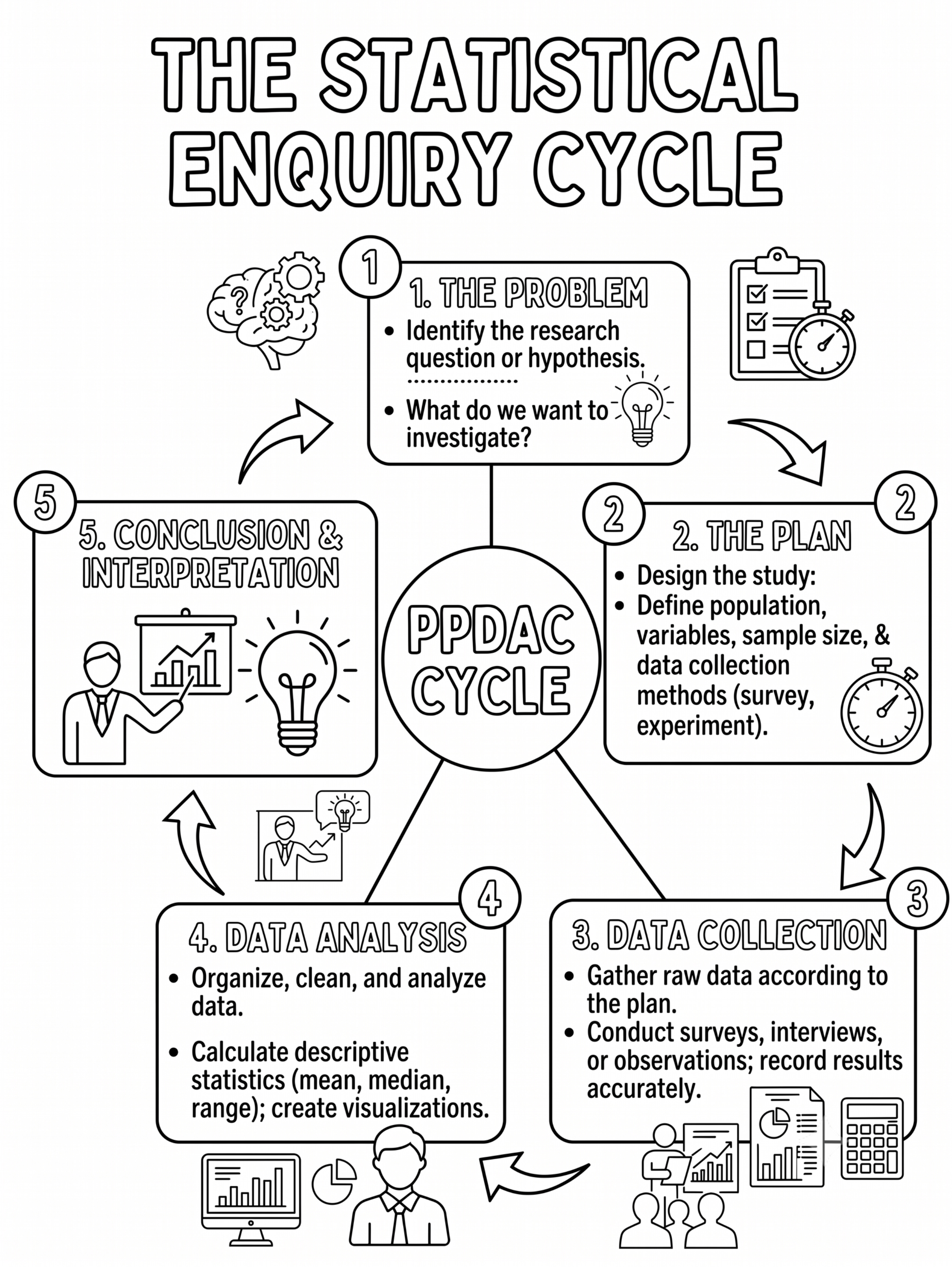

Figure 4.1.: Statistical enquiry cycle. Fill in the blanks and colour away your worries.

New Zealand's statistics curriculum uses the **PPDAC cycle** as the framework for carrying out statistical investigations and interpreting statistical information.

Stage	What you do
Problem	Pose a statistical question
Plan	Plan how to collect data
Data	Collect and record data
Analysis	Explore and summarise the data
Conclusion	Interpret results and answer the question

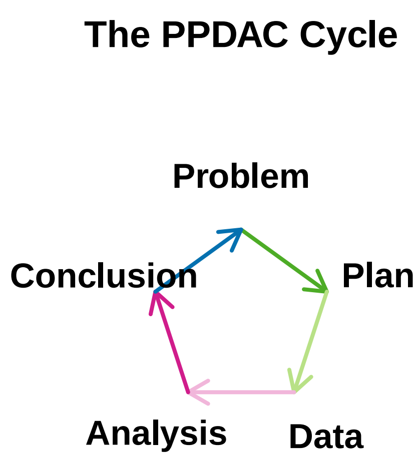

Figure 4.2.: The PPDAC cycle: a framework for statistical investigations.

Why PPDAC matters in both standards

- In **AS1.1**, you carry out the full cycle yourself.
- In **AS1.3**, you often interpret someone else's data and reasoning.
- In both cases, you still need a clear question, evidence, and a conclusion in context.

4.1. Stage-by-stage guidance

4.1.1. Problem

Write a statistical question that expects variability and identifies a clear context, variable, and population.

4.1.2. Plan

Decide how data will be collected, how the sample will be selected, and how fairness and bias will be managed.

4.1.3. Data

Collect and record the data accurately, with units and labels that can be interpreted later.

4.1.4. Analysis

Use graphs and summary statistics to identify patterns, centre, spread, and interesting features.

4.1.5. Conclusion

Answer the original question in context, using evidence from the analysis and acknowledging any limitations.

4.2. Common PPDAC mistakes

- Starting analysis before writing a clear statistical question.
- Using a sample that is large but biased.
- Reporting numbers without units or context.
- Giving a conclusion that does not answer the original question.

4.3. Quick check

1. Which PPDAC stage is most closely linked to reducing sampling bias?
2. Which stage should include comparison language and evidence from graphs?
3. Why must the conclusion link back to the original question?

Answers

1. **Plan**.
2. **Analysis** (with final interpretation carried into **Conclusion**).
3. Because PPDAC is question-driven, so conclusions must directly answer the statistical problem in context.

5. Types of Data and Variables

Understanding the **type of data** you have is the first step in choosing the right graph or summary statistic. The wrong choice can make data hard to understand — or even misleading.

5.1. Categorical vs numerical data

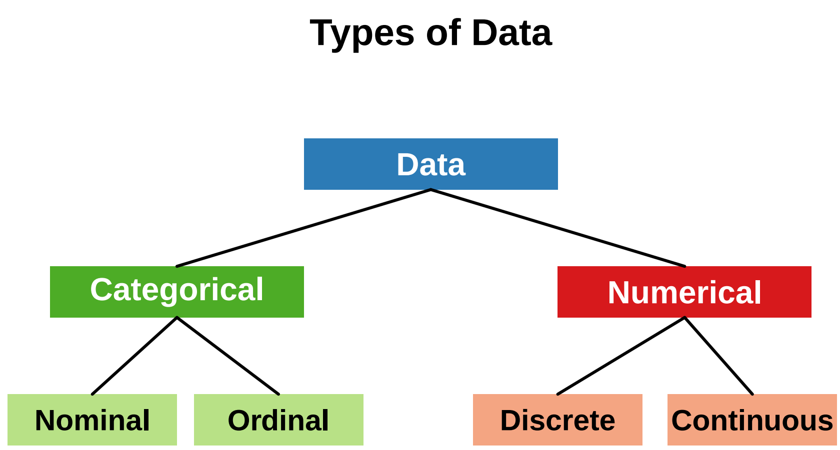

Figure 5.1.: Classification of data types.

5.2. Categorical data

Categorical (qualitative) data describes qualities or categories. The values are labels, not numbers.

5.2.1. Nominal data

Nominal data has **no natural order**. Categories are just names.

Examples: - Eye colour (brown, blue, green, grey) - Favourite sport (rugby, netball, football, basketball) - Type of transport to school (bus, car, walk, bike)

5.2.2. Ordinal data

Ordinal data has a **natural order**, but the gaps between categories are not equal.

Examples: - Survey ratings (strongly disagree → disagree → neutral → agree → strongly agree) - Year group (Year 9, Year 10, Year 11, Year 12, Year 13) - Clothing size (XS, S, M, L, XL)

5.3. Numerical data

Numerical (quantitative) data represents quantities that can be measured or counted.

5.3.1. Discrete data

Discrete data can only take **specific, separate values** (usually whole numbers from counting).

Examples: - Number of siblings - Number of text messages sent per day - Goals scored in a rugby match

5.3.2. Continuous data

Continuous data can take **any value within a range** (usually from measuring).

Examples: - Height (cm) - Weight (kg) - Time to complete a race (seconds) - Temperature (°C)

5.4. Real data example

Table 5.1.: Example student survey data — identify the type of each variable.

Student	Gender	Height_cm	Siblings	Transport	Rating
S1	Female	154.3	1	Bus	Fair
S2	Non-binary	144.2	3	Bike	Excellent
S3	Non-binary	163.2	2	Bus	Excellent
S4	Non-binary	162.6	3	Bus	Poor
S5	Female	181.2	0	Car	Excellent
S6	Non-binary	152.9	2	Bike	Fair
S7	Female	167.8	3	Bus	Good
S8	Female	164.5	0	Walk	Excellent
S9	Non-binary	161.7	2	Bus	Fair
S10	Female	159.0	4	Walk	Fair

Identify the variable types

- **Gender** → Categorical, Nominal
- **Height_cm** → Numerical, Continuous
- **Siblings** → Numerical, Discrete
- **Transport** → Categorical, Nominal
- **Rating** → Categorical, Ordinal

5.5. Choosing the right display

The type of data determines the best graph to use:

Data type	Suitable displays
Categorical (nominal/ordinal)	Bar chart, pie chart, dot plot
Numerical (discrete)	Dot plot, bar chart, stem-and-leaf
Numerical (continuous)	Histogram, box plot, dot plot, stem-and-leaf
Two numerical variables	Scatter plot

5.6. Glossary

Term	Meaning
Variable	A characteristic that varies between individuals
Categorical	Data that falls into named groups or categories
Nominal	Categorical with no natural order
Ordinal	Categorical with a natural order
Numerical	Data that can be measured or counted
Discrete	Numerical data that can only take specific values
Continuous	Numerical data that can take any value in a range

5.7. Check your understanding

Classify each variable as **categorical (nominal/ordinal)** or **numerical (discrete/continuous)**:

1. Shoe size
2. Favourite subject at school
3. Time spent sleeping (hours per night)
4. Number of books read last month
5. Blood type
6. Temperature in a classroom
7. Year level
8. Distance from home to school (km)

Answers

1. Numerical, discrete (or ordinal — shoe sizes have a natural order but are not continuous)
2. Categorical, nominal
3. Numerical, continuous
4. Numerical, discrete
5. Categorical, nominal
6. Numerical, continuous
7. Categorical, ordinal (or numerical discrete depending on context)
8. Numerical, continuous

6. Statistical Graphs and Displays

A well-chosen graph makes patterns in data immediately visible. In this chapter you will learn to **create**, **read**, and **critically interpret** the most common statistical displays.

6.1. Setting up our data

Throughout this chapter we use a simulated survey of 60 Year 11 students across a New Zealand school. The data includes variables on transport, height, study time, and maths scores.

6.2. Bar charts

Bar charts display the **frequency** (count) or **relative frequency** (proportion) of categorical data.

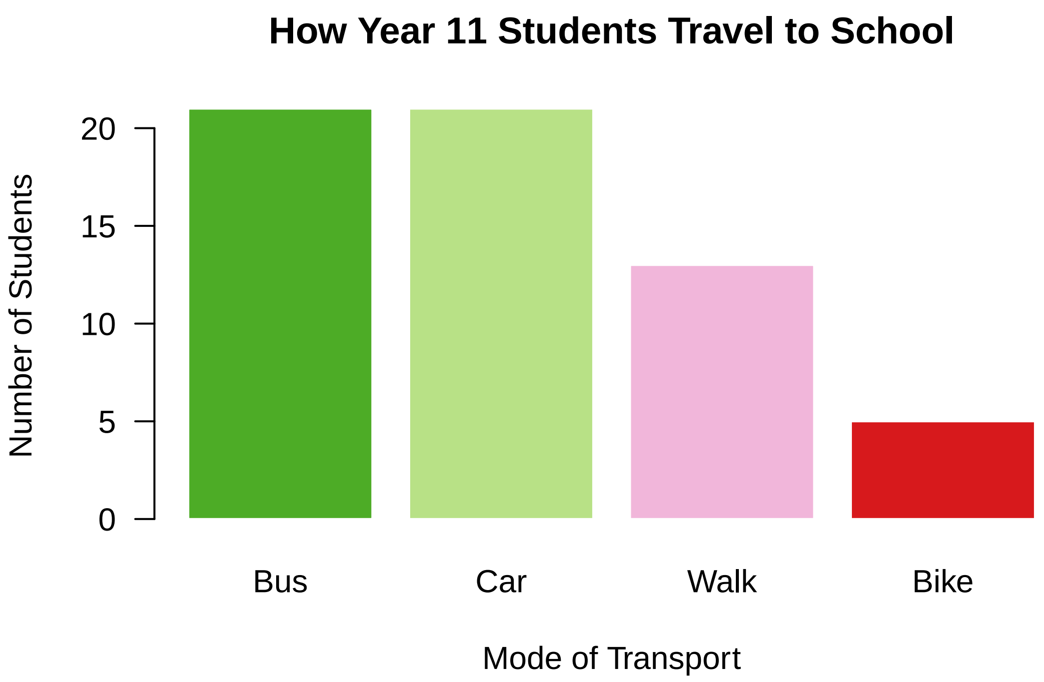

Figure 6.1.: Bar chart showing how Year 11 students travel to school.

6.2.1. Reading a bar chart

Tip

When describing a bar chart, always comment on:

1. **The most and least common** categories
2. **The overall pattern** (are categories roughly equal, or is one dominant?)
3. **The context** — what does this tell us about the group?

Sample answer: "Bus is the most common way students travel to school, with over 20 students using this method. Bike is the least common. More students use public or active transport (bus, walk, bike) than passive transport (car)."

6.3. Dot plots

Dot plots are useful for **small numerical datasets**. Each data point is shown as a dot above a number line.

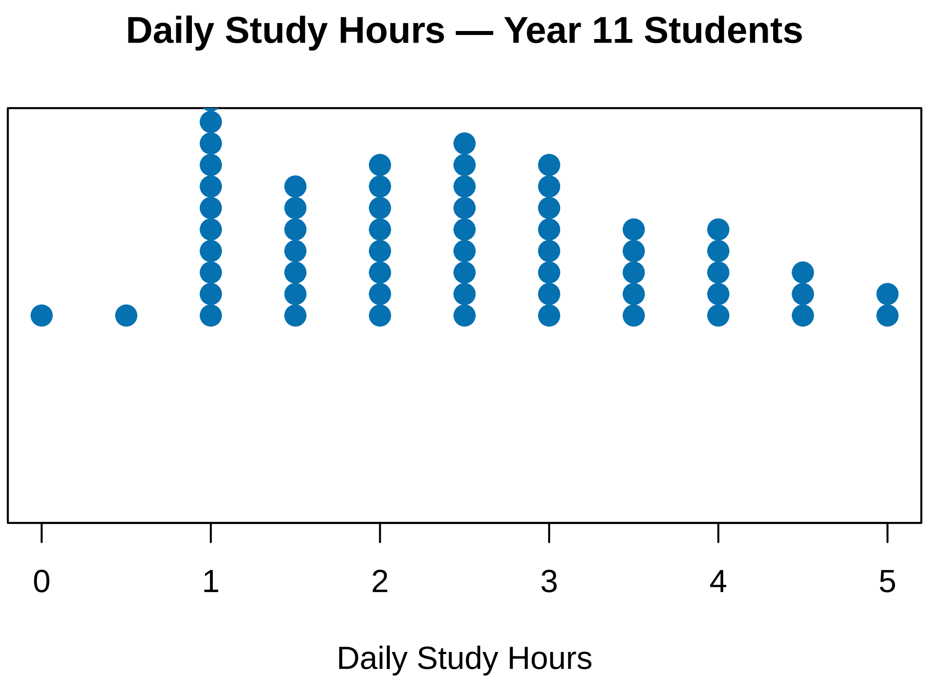

Figure 6.2.: Dot plot of daily study hours for Year 11 students.

6.4. Histograms

A histogram shows the **distribution of continuous numerical data** by grouping values into intervals (bins).

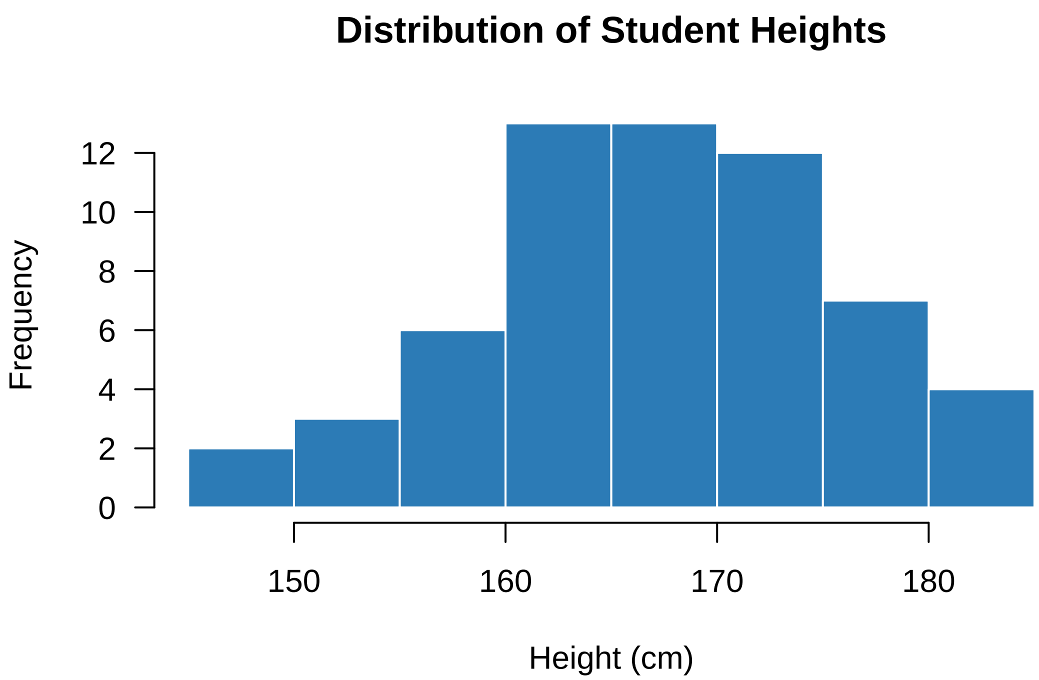

Figure 6.3.: Histogram of student heights (cm).

6.4.1. Describing a histogram — SOCS

Use **SOCS** to describe a distribution:

Letter	What to describe
Shape	Symmetric, skewed left, skewed right, uniform, bimodal
Outliers	Any unusual values far from the main group
Centre	Where the middle of the data is (mean or median)
Spread	How variable the data is (range, IQR, standard deviation)

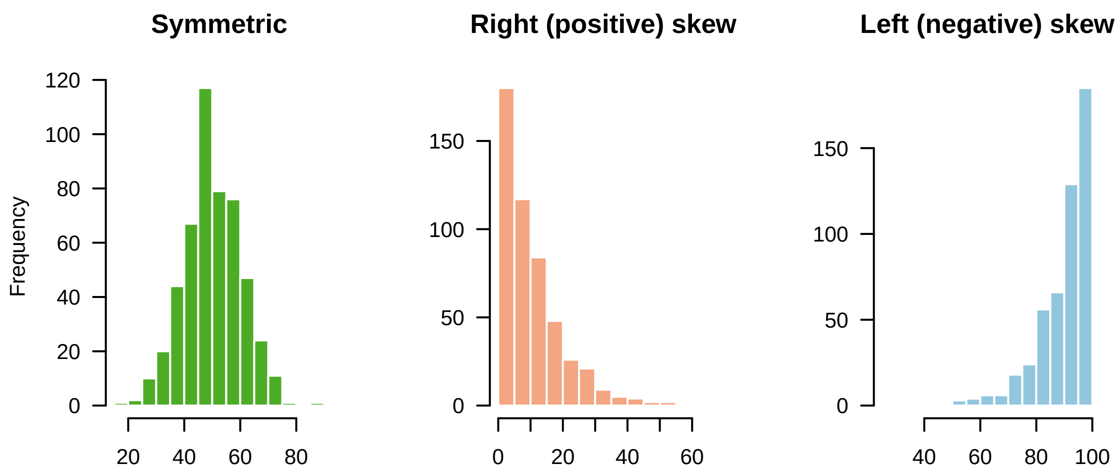

Figure 6.4.: Examples of different distribution shapes.

6.5. Box plots (box-and-whisker plots)

A box plot summarises a **five-number summary**: Minimum, Q1, Median, Q3, Maximum

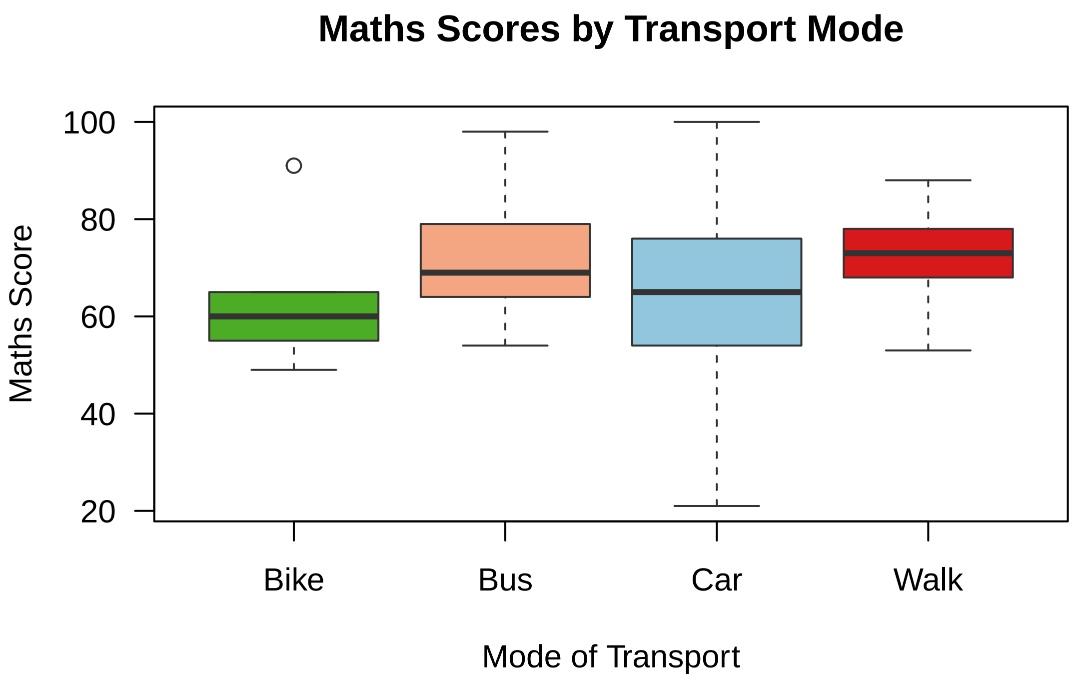

Figure 6.5.: Box plot of maths scores by transport type.

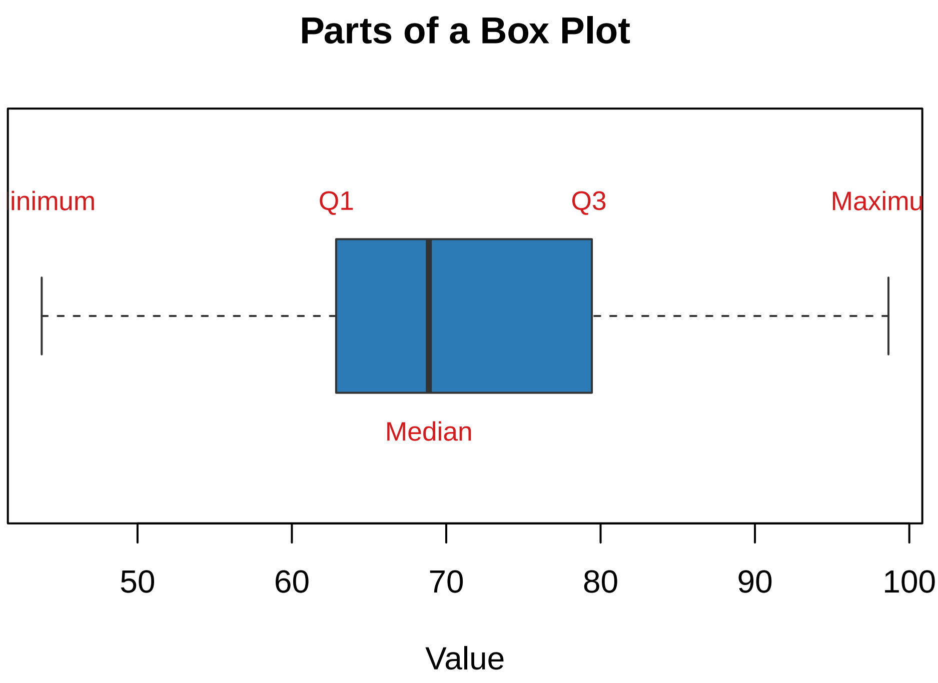

Figure 6.6.: Anatomy of a box plot.

```
#>     6 | 04557789
#>     7 | 0169
#>     8 | 23448
#>     9 | 18
#>    10 | 0
```

6.7. Scatter plots

Scatter plots show the **relationship between two numerical variables**.

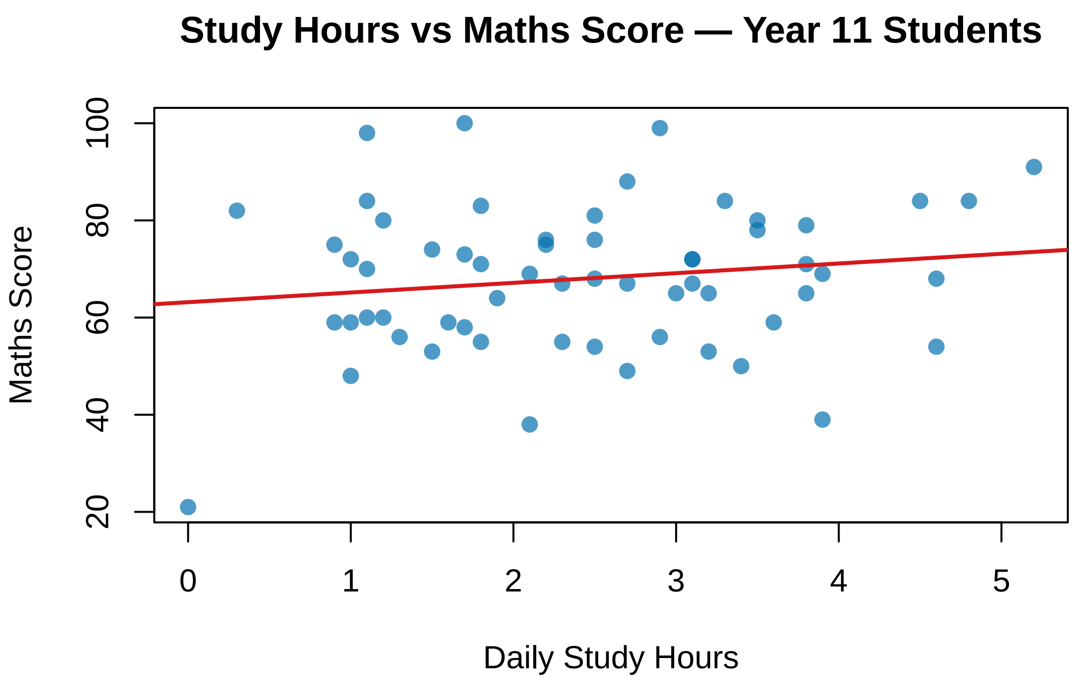

Figure 6.7.: Scatter plot of study hours vs maths score.

6.7.1. Describing a scatter plot

Comment on:

1. **Direction** — positive, negative, or no association
2. **Form** — linear or non-linear
3. **Strength** — strong, moderate, or weak
4. **Outliers** — any unusual points

Correlation Causation

Just because two variables are related **does not** mean one causes the other. Always think about whether an association makes sense in context.

6.8. Pie charts

Pie charts show **proportions** of categorical data. Use them carefully — bar charts are usually easier to read precisely.

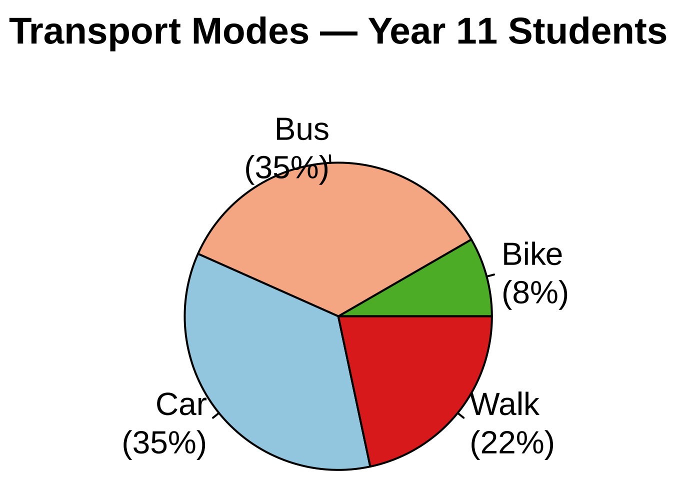

Figure 6.8.: Pie chart of transport modes.

6.9. Choosing the right graph

Situation	Best graph
One categorical variable	Bar chart, pie chart
One numerical variable (small n)	Dot plot, stem-and-leaf
One numerical variable (large n)	Histogram, box plot
Comparing groups numerically	Side-by-side box plots, back-to-back stem-and-leaf
Two numerical variables	Scatter plot

6.10. Check your understanding

1. A student recorded the heights of 25 classmates. Which graph type is most appropriate to display this data? Explain your choice.
2. Describe the scatter plot in Figure 6.7 using the four features (direction, form, strength, outliers).
3. A bar chart shows the following transport counts: Bus = 24, Car = 15, Walk = 15, Bike = 6. What percentage of students walk to school?

4. True or false: “A right-skewed distribution has a longer tail on the left.” Explain.

Answers

1. A histogram or dot plot — height is numerical and continuous.
2. Positive direction (more study → higher score), roughly linear form, moderate strength, possibly one or two outliers.
3. Walk = 15/60 = 25%.
4. False — a right-skewed (positively skewed) distribution has a longer tail on the RIGHT.

7. Bivariate Data in Ecology

Investigating koura habitats using the PPDAC cycle

In this chapter, we apply the PPDAC (Problem, Plan, Data, Analysis, Conclusion) statistical enquiry cycle to a real-world ecological context. Understanding how two numerical variables interact is a core component of NCEA Level 1 Statistics.

7.1. Exercise: Project Kōura

Context: A group of students—Jack, Trishav, and Levi—are conducting a local ecological survey, which they have named "Project Kōura". They are investigating the populations of native freshwater crayfish (kōura) across several local catchments, including the Kaikorai Stream and the Water of Leith. They want to know if the physical environment of the stream affects the growth of the kōura. They have accessed an open dataset from the New Zealand Freshwater Fish Database (NZFFD) containing recent survey results.

7.1.1. 1. Problem

Before looking at the data, the group needs to define a clear investigative question. Jack suggests they look at the relationship between the water temperature of the stream and the size of the kōura.

Task A: Write a suitable investigative question for this survey. Ensure you specify the variables, the population, and the direction of your investigation.

Task B: Write a hypothesis. Based on what you know about freshwater biology, what kind of relationship do you expect to see between water temperature and kōura carapace length?

7.1.2. 2. Plan

The students isolate two numerical variables from the NZFFD dataset:

- **Explanatory Variable:** Stream Temperature (°C)
- **Response Variable:** Carapace Length (mm)

Task C: Why is Stream Temperature the explanatory variable and Carapace Length the response variable, rather than the other way around?

7.1.3. 3. Data

We will simulate importing the NZFFD dataset directly into R to prepare it for analysis.

```
#> # A tibble: 6 x 3
#>   catchment       temp_c carapace_length_mm
#>   <chr>            <dbl>              <dbl>
#> 1 Taieri River     11.0               37.1
#> 2 Taieri River     15.9               36.0
```

```
#> 3 Taieri River      9.23                48.2
#> 4 Kaikorai Stream   8.73                48.1
#> 5 Taieri River      9.14                46.7
#> 6 Kaikorai Stream  13.5                 37.2
```

8. Measures of Centre and Spread

Once you have a graph, you need **numerical summaries** to describe and compare distributions precisely. The two key things to describe are:

- **Centre** — where is the "middle" of the data?
- **Spread** — how variable is the data?

8.1. Measures of centre

8.1.1. The mean

The **mean** (average) is calculated by adding all values and dividing by the number of values:

$$\bar{x} = \frac{\sum x_i}{n}$$

```
#> Scores: 52 55 60 62 66 68 70 71 72 73 74 75 78 79 80 83 85 88 91 95
#> Mean:   73.85
#> n =     20
```

8.1.2. The median

The **median** is the middle value when data is sorted from smallest to largest. It is not affected by outliers.

- If n is **odd**: median = middle value
- If n is **even**: median = average of the two middle values

```
#> Sorted scores:
#>  52 55 60 62 66 68 70 71 72 73 74 75 78 79 80 83 85 88 91 95
#> n = 20 (even) → median = mean of 10th and 11th values
#> 10th value: 73
#> 11th value: 74
#> Median = 73.5
```

8.1.3. The mode

The **mode** is the most frequently occurring value. Data can have: - **No mode** (all values appear once) - **One mode** (unimodal) - **Two modes** (bimodal) - **More than two modes** (multimodal)

```
#> Data: 4 7 2 7 9 7 3 4 5 4
#> Mode(s): 4 7
```

8.1.4. When to use each measure of centre

Measure	Best used when...	Affected by outliers?
Mean	Data is roughly symmetric, no outliers	Yes
Median	Data is skewed or has outliers	No
Mode	Data is categorical; or finding most common value	No

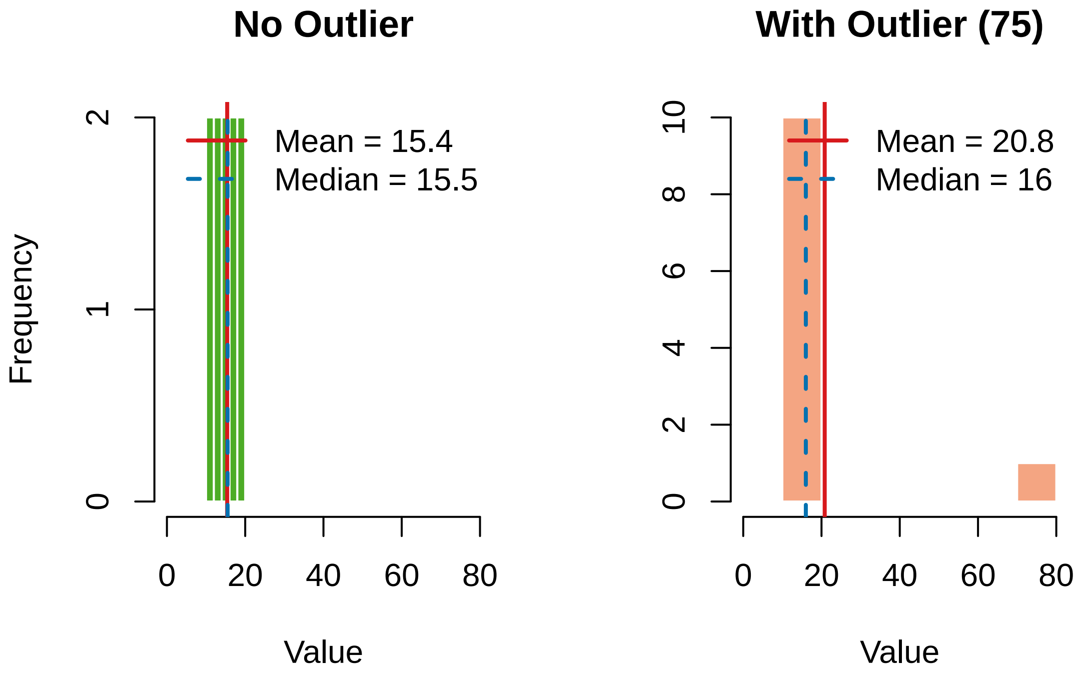

Figure 8.1.: Effect of outliers on mean vs median.

8.2. Measures of spread

8.2.1. Range

The **range** is the simplest measure of spread:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

```
#> Range of scores: 43
#>   (Max = 95 , Min = 52 )
```

8.2.2. Quartiles and the interquartile range (IQR)

Quartiles divide sorted data into four equal groups:

Quartile	Description
Q1 (lower quartile)	25% of data falls below this value
Q2 (median)	50% of data falls below this value
Q3 (upper quartile)	75% of data falls below this value

The **interquartile range (IQR)** is the spread of the middle 50% of data:

$$\text{IQR} = Q3 - Q1$$

```
#> Five-number summary:
#>   Minimum: 52
#>   Q1:      67.5
#>   Median:  73.5
#>   Q3:      80.75
#>   Maximum: 95
#>   IQR = 13.25
```

8.2.3. Standard deviation

The **standard deviation** measures how spread out data values are from the mean. A larger standard deviation means more variability.

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$$

```
#> Mean: 73.85
#> Standard deviation: 11.52
```

> **Rule of thumb**
>
> For roughly symmetric distributions, about **68%** of data falls within one standard deviation of the mean, and **95%** falls within two standard deviations (the empirical rule / 68-95-99.7 rule).

8.2.4. Choosing the right measure of spread

Measure	Use with...	Advantage
Range	Quick overview	Easy to calculate
IQR	Skewed data or when outliers present	Resistant to outliers
Standard deviation	Symmetric data, no outliers	Uses all data values

8.3. Comparing groups

A common task in AS91946 is to **compare two or more groups**. Always compare both centre AND spread.

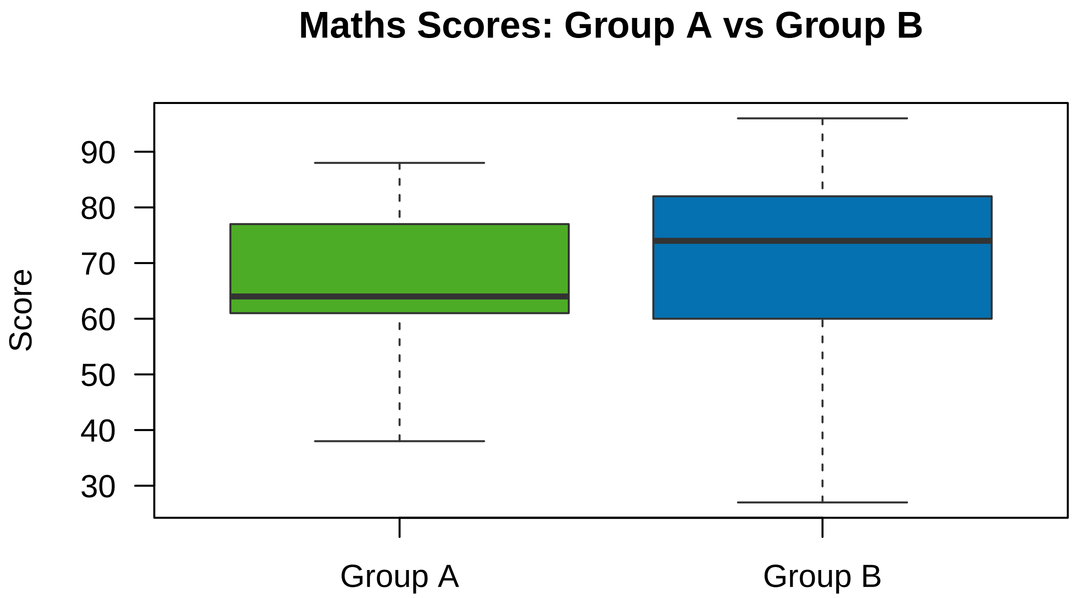

Figure 8.2.: Comparing maths scores between Group A and Group B.

Table 8.4.: Summary statistics for Groups A and B

Group	n	Mean	Median	SD	IQR	Min	Max
Group A	30	65.6	64	12.6	14.25	38	88
Group B	30	70.2	74	15.8	21.25	27	96

8.3.1. Model comparison paragraph (Merit level)

> "Group B has a higher median score (72) than Group A (65), suggesting that Group B performed better overall. However, Group B also has a larger IQR (approximately 20 vs 14) and standard deviation (15.1 vs 9.8), indicating their scores were more variable. Group A's performance was more consistent. Both groups appear to have roughly symmetric distributions with no obvious outliers."

8.4. Identifying outliers

A value is often flagged as an **outlier** if it falls more than 1.5 × IQR below Q1 or above Q3:

$$\text{Lower fence} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper fence} = Q3 + 1.5 \times \text{IQR}$$

```
#> Q1 = 67.5
#> Q3 = 80.75
#> IQR = 13.25
```

```
#> Lower fence: 47.625
#> Upper fence: 100.625
#> Outliers:
```

8.5. Glossary

Term	Meaning
Mean	Sum of all values divided by the count
Median	Middle value when data is sorted
Mode	Most frequently occurring value
Range	Maximum − Minimum
IQR	Q3 − Q1; spread of the middle 50%
Standard deviation	Average distance of values from the mean
Outlier	A value that lies far from the rest of the data
Five-number summary	Min, Q1, Median, Q3, Max

8.6. Check your understanding

Use the dataset: **12, 15, 18, 18, 20, 22, 25, 25, 28, 100**

1. Calculate the mean and median. Which is more appropriate as a measure of centre for this data? Why?
2. Find Q1, Q3, and the IQR.
3. Calculate the range.
4. Is 100 an outlier? Use the 1.5 × IQR rule to justify your answer.
5. Sketch a box plot of this data.

Answers

1. Mean $= (12{+}15{+}18{+}18{+}20{+}22{+}25{+}25{+}28{+}100)/10 = 283/10 = 28.3$; Median $= (20{+}22)/2 = 21$. The **median** is more appropriate because of the outlier (100) which pulls the mean up.
2. $Q1 = 18$, $Q3 = 25$, $IQR = 7$.
3. Range $= 100 - 12 = 88$.
4. Upper fence $= 25 + 1.5 \times 7 = 25 + 10.5 = 35.5$. Since $100 > 35.5$, **yes**, 100 is an outlier.
5. Sketch: whiskers from 12 to 35.5, box from 18 to 25, median line at 21, and a dot at 100 representing the outlier.

9. Practice Questions and Achievement Standard Tips

This chapter brings everything together with practice questions at each grade level, and advice on how to approach the AS91946 assessment.

9.1. About the achievement standard

Detail	Information
Standard number	AS91946
Title	Interpret and apply mathematical and statistical information in context
Level	NCEA Level 1
Credits	4
Assessment	External (end-of-year exam)
Strands	Statistics and Probability

9.2. What each grade requires

9.2.1. Achieved

- Correctly read values from graphs and tables
- Calculate basic statistics (mean, median, mode, range)
- State probabilities from given information
- Identify variable types
- Make simple statements about data in context

9.2.2. Merit

- **Justify** your answers with reasons
- Compare and contrast groups using both centre AND spread
- Identify potential sources of bias
- Explain the difference between correlation and causation
- Interpret probabilities and relate them to real-world context

9.2.3. Excellence

- Show **statistical insight** — go beyond the obvious
- Consider alternative explanations or limitations
- Make connections between different parts of the data
- Critically evaluate statistical claims
- Use appropriate statistical language precisely

9.3. Writing tips for the exam

> The golden rule
>
> **Always refer back to the context.** Every answer should mention: - The **variable** being described - The **units** of measurement - The **group** being described (population/sample)
> “The mean is 65.”
> “The mean maths score for Year 11 students was 65 out of 100.”

9.3.1. Comparison sentence starters

Purpose	Sentence starters
Centre	“On average…”; “The typical…”; “The median…is higher/lower than…”
Spread	“The scores are more/less variable…”; “The IQR is larger, suggesting…”
Shape	“The distribution is roughly symmetric / right-skewed…”
Outliers	“There appears to be an outlier at…which may indicate…”
Causation	“However, this may be due to…rather than a direct causal relationship”

9.4. Practice set 1 — Achieved level

```
#> Heights (cm) of 20 Year 11 students:
#> 147.3 153.9 154.5 157.3 158.9 159.5 163.6 164.3 164.8 165 165.2 166.4 166.5 167.2 169.3 170.1
```

Questions:

1. What type of data is height? Give a reason.
2. Calculate the mean height. Round to one decimal place.
3. Find the median height.
4. Find the range.
5. A student drew a histogram of this data. Circle the most appropriate description: **symmetric / right-skewed / left-skewed**

> Worked solutions
>
> 1. Numerical, continuous — height can take any value within a range.
> 2. Mean: 164.8 cm
> 3. Median: 165.1 cm
> 4. Range: 37.3 cm
> 5. Likely symmetric (heights tend to be normally distributed).

9.5. Practice set 2 — Merit level

A local council surveyed 80 residents about their weekly exercise hours. The results were:

Table 9.3.: Weekly exercise hours: adults vs teenagers

Group	n	Mean	Median	SD	IQR	Min	Max
Adults	40	4.45	4.35	2.07	2.85	0	7.9
Teenagers	40	6.58	7.10	2.85	3.58	0	13.0

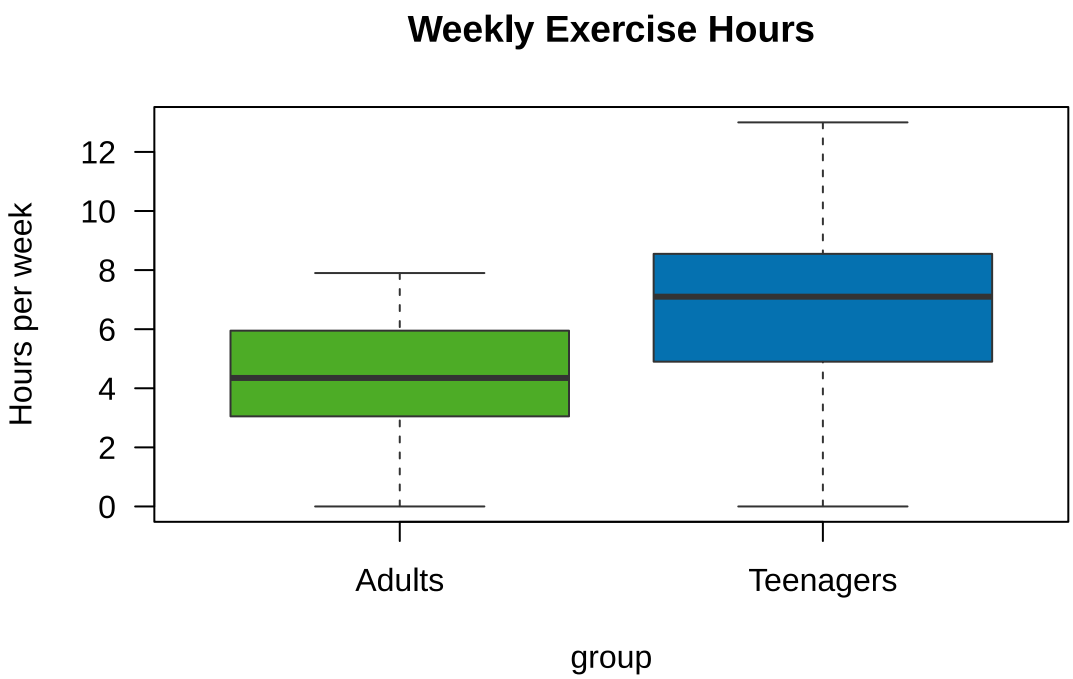

Figure 9.1.: Weekly exercise hours for adults and teenagers.

Questions:

1. Compare the centre of the two distributions. What does this tell us about exercise habits in context?
2. Compare the spread. What does this tell us?
3. The council used a telephone survey conducted at 2 pm on weekdays. Identify one source of bias and explain how it might affect the results.
4. Can we conclude from this data that teenagers exercise more because they are younger? Explain.

ℹ Model answers

1. Centre: The median exercise hours for teenagers (approx. 7.1 hrs) is higher than for adults (4.35 hrs). This suggests teenagers in this sample tend to do more weekly exercise than adults.
2. Spread: The IQR and standard deviation are larger for teenagers (3.6 hrs vs 2.9 hrs). This means teenagers' exercise habits are more variable — some exercise a lot while others

exercise very little. Adults' exercise habits are more consistent.
3. Bias: Calling at 2 pm on weekdays means working adults are unlikely to answer (they are at work). This leads to **non-response bias** — the adult sample may over-represent retired or part-time workers, who may exercise more than the general adult population.
4. Causation: No — we cannot conclude that age *causes* the difference in exercise. Teenagers may exercise more because they participate in school sport, not because of their age per se. Age and exercise are *correlated* in this sample, but there may be confounding variables such as school PE requirements, free time, or lifestyle factors.

9.6. Practice set 3 — Excellence level

A researcher claims: *"A study of 50 Auckland households found that households with more bookshelves have higher household income. Therefore, buying more bookshelves will increase your income."*

Questions:

1. Identify the two variables and their types.
2. Explain why this conclusion is flawed. Use statistical language.
3. Suggest one confounding variable that could explain the relationship.
4. What would the researcher need to do to properly establish causation?
5. The sample was 50 Auckland households selected by going door-to-door in Remuera (a wealthy suburb). Evaluate the reliability of generalising these findings to all New Zealand households.

Model answers

1. Number of bookshelves: numerical, discrete. Household income: numerical, continuous.
2. The conclusion confuses **correlation with causation**. Even if the two variables are positively correlated, this does not mean bookshelves cause higher income. The relationship could be due to a third variable, and correlation studies cannot establish cause-and-effect without experimental control.
3. Wealth/education level — wealthier, more educated households are more likely to own books/bookshelves AND to have higher income. Both variables are driven by a common factor.
4. To establish causation, the researcher would need a **controlled experiment** — randomly assigning households to receive bookshelves or not, holding all other variables constant, and measuring income change over time. This would be impractical.
5. The sample has significant **sampling bias**. Remuera is one of Auckland's wealthiest suburbs. The sample does not represent the diversity of New Zealand households in terms of income, region, or ethnicity. Results cannot be reliably generalised to all New Zealand households. A **stratified random sample** from across New Zealand would be needed.

9.7. Key formulas summary

Measure	Formula
Mean	$\bar{x} = \Sigma x \ / \ n$
Range	Max − Min

Measure	Formula
IQR	Q3 − Q1
Probability	P(A) = favourable outcomes / total outcomes
Complement	P(A') = 1 − P(A)
Lower outlier fence	Q1 − 1.5 × IQR
Upper outlier fence	Q3 + 1.5 × IQR

9.8. Exam strategy checklist

Before the exam, make sure you can:

- ☐ Identify variable types (categorical/numerical, nominal/ordinal, discrete/continuous)
- ☐ Read values accurately from bar charts, histograms, box plots, and scatter plots
- ☐ Calculate mean, median, mode, range, IQR
- ☐ Describe distributions using SOCS (Shape, Outliers, Centre, Spread)
- ☐ Compare two groups using both centre and spread in context
- ☐ Calculate simple probabilities and use complementary events
- ☐ Read and calculate from two-way tables
- ☐ Identify bias in data collection methods
- ☐ Distinguish correlation from causation
- ☐ Critically evaluate statistical claims in media

9.9. Useful resources

Resource	Description	URL
NZQA	Official achievement standard information	nzqa.govt.nz
CensusAtSchool	Real NZ student data for practice	new.censusatschool.org.nz
iNZight	Free NZ-made statistics software	inzight.nz
Stats NZ	Official NZ statistics and data	stats.govt.nz
Khan Academy	Free video lessons on statistics	khanacademy.org

Part II.

Part 2: Probability and Critical Reading

10. Probability

Probability is a measure of how likely an event is to occur. It is expressed as a number between 0 (impossible) and 1 (certain), or as a percentage.

$$P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}$$

10.1. Probability scale

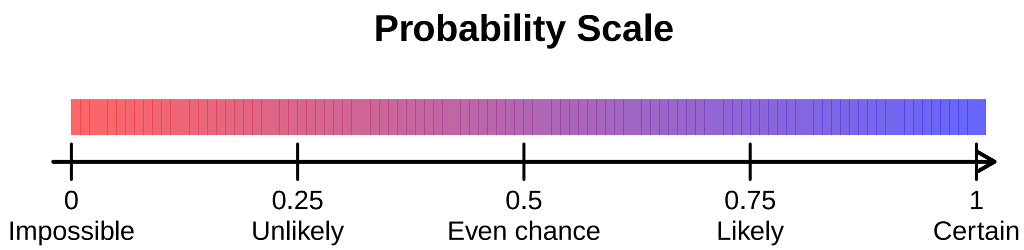

Figure 10.1.: The probability scale from impossible (0) to certain (1).

10.2. Key vocabulary

Term	Definition
Experiment	A process with uncertain outcomes
Outcome	A single possible result
Event	One or more outcomes grouped together
Sample space	The set of all possible outcomes
Complementary event	All outcomes that are NOT in the event

10.3. Simple probability — equally likely outcomes

When all outcomes are **equally likely**, we can use the classical definition.

10.3.1. Example: rolling a die

```
#> Sample space: 1 2 3 4 5 6
#> P(rolling a 4) = 0.1666667
#> P(even number) = 0.5
#> P(> 4) = 0.3333333
```

10.4. Complementary events

The **complement** of event A (written A' or $\bar{A}$) contains all outcomes NOT in A.

$$P(A') = 1 - P(A)$$

Example: If the probability of rain tomorrow is 0.3, then:

$$P(\text{no rain}) = 1 - 0.3 = 0.7$$

```
#> P(rain) = 0.3
#> P(no rain) = 1 - 0.3 = 0.7
```

10.5. Experimental (relative frequency) probability

When we cannot calculate probability theoretically, we **estimate** it by repeating an experiment many times:

$$P(\text{event}) \approx \frac{\text{number of times event occurred}}{\text{total number of trials}}$$

10.5.1. Simulating coin flips

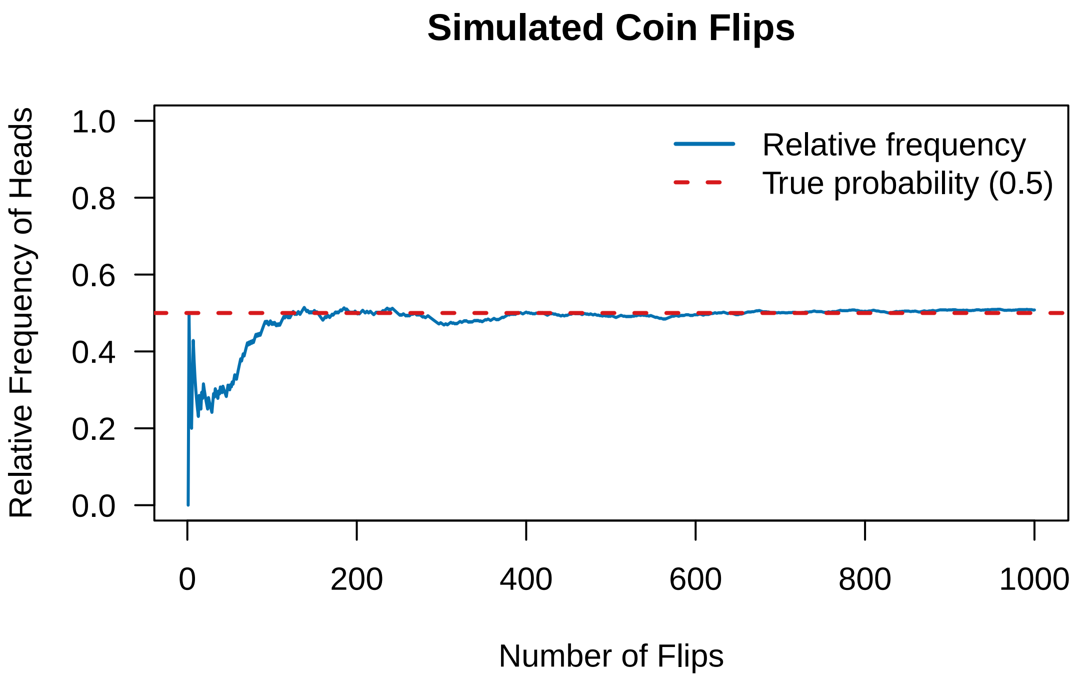

Figure 10.2.: Relative frequency of heads approaches 0.5 as the number of flips increases (the law of large numbers).

> **Law of Large Numbers**
>
> As the number of trials increases, the **relative frequency** (experimental probability) gets closer and closer to the **true probability**. This is why surveys and experiments need large samples.

10.6. Two-way tables

A **two-way table** (contingency table) organises data by two categorical variables and is very useful for calculating probabilities.

Table 10.2.: Sport preference by gender (n = 120)

	Female	Male	Sum
Basketball	16	11	27
Football	13	17	30
Netball	18	17	35
Rugby	16	12	28
Sum	63	57	120

10.6.1. Reading probabilities from a two-way table

Using the table above:

```
#> P(Male) = 57 / 120 = 0.475
#> P(Rugby) = 28 / 120 = 0.233
#> P(Male AND Rugby) = 12 / 120 = 0.1
#> P(Rugby | Male) = 12 / 57 = 0.211
```

The last probability — P(Rugby | Male) — is a **conditional probability**: the probability of preferring rugby, **given** the student is male.

10.7. Tree diagrams

Tree diagrams help calculate probabilities for **sequential events**.

Key rule: Multiply probabilities along each branch, add probabilities for different paths leading to the same outcome.

10.8. Expected value (extension)

For a simple scenario, the **expected value** is the long-run average outcome:

$$E(X) = \sum x \cdot P(x)$$

Example: A raffle has 200 tickets at $2 each. First prize is $100, second prize is $50. What is the expected gain per ticket?

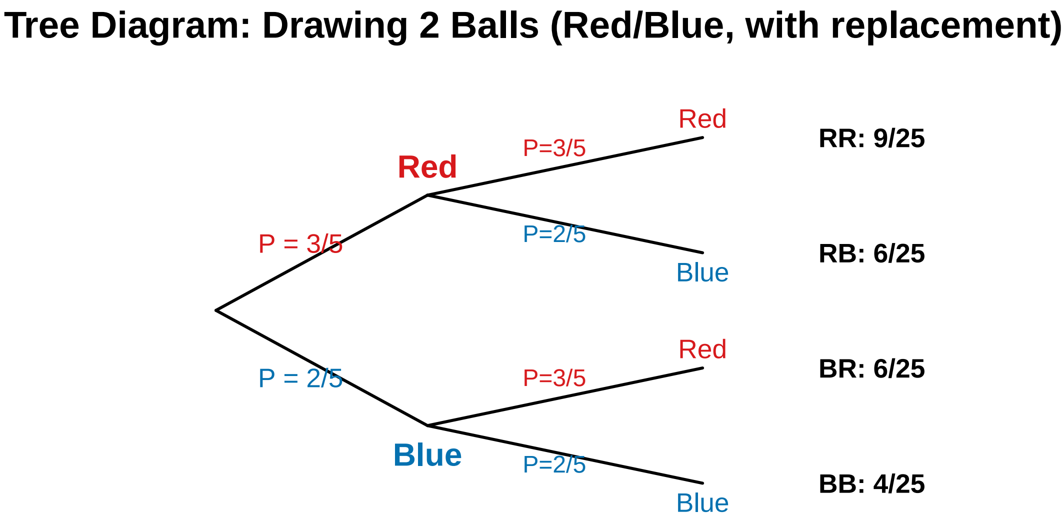

Figure 10.3.: Tree diagram for drawing two balls from a bag (with replacement).

```
#> Expected winnings: 0.75
#> Ticket cost:       2
#> Expected net gain: -1.25
#> (A negative expected gain means this is not a good investment!)
```

10.9. Glossary

Term	Definition
Probability	Likelihood of an event, from 0 to 1
Sample space	All possible outcomes
Event	One or more outcomes
Complement	All outcomes NOT in the event
Relative frequency	Experimental estimate of probability
Conditional probability	P(A given B): probability of A given B occurred
Two-way table	Table showing two categorical variables together
Tree diagram	Diagram showing sequential outcomes and probabilities

10.10. Check your understanding

1. A bag contains 4 red, 3 blue, and 3 yellow marbles. A marble is drawn at random. Find:
   a. P(red)
   b. P(not red)
   c. P(blue or yellow)

2. A class of 30 students was surveyed: 18 play sport, 12 play an instrument, and 6 do both.
   a. Draw a Venn diagram.
   b. Find P(plays sport but not instrument).
   c. Find P(neither sport nor instrument).
3. If $P(A) = 0.4$ and $P(B) = 0.3$ and A and B are independent, find P(A and B).

Answers

1. a. $4/10 = 0.4$; b. $6/10 = 0.6$; c. $6/10 = 0.6$
2. b. $12/30 = 0.4$; c. $(30 - 18 - 12 + 6)/30 = 6/30 = 0.2$
3. $P(\text{A and B}) = 0.4 \times 0.3 = 0.12$ (for independent events)

11. Interpreting Statistical Information

A major focus of AS91946 is the ability to **critically read and interpret** statistical information from a variety of sources — including newspapers, websites, government reports, and scientific studies.

11.1. Reading statistical claims

Statistical claims appear everywhere in everyday life. As a critical consumer of information, you need to ask:

1. **Who** collected the data and why?
2. **How** was the data collected (survey, experiment, census)?
3. **How many** people or things were measured (sample size)?
4. **Who** was sampled — does the sample represent the population?
5. **What** do the numbers actually mean in context?

11.2. Sources of bias

Bias is a systematic error that causes results to be unrepresentative.

Type of bias	Description	Example
Sampling bias	Sample does not represent population	Surveying only Year 11s about a school-wide issue
Response bias	People don't answer truthfully	"Do you cheat on tests?"
Non-response bias	Certain groups are less likely to respond	Online survey misses those without internet
Leading question bias	Question wording influences the answer	"Don't you agree that homework is too much?"
Volunteer bias	Only motivated people respond	Self-selected online polls

11.3. Sample size and reliability

Larger samples generally give more **reliable** results — but size alone is not enough if the sample is biased.

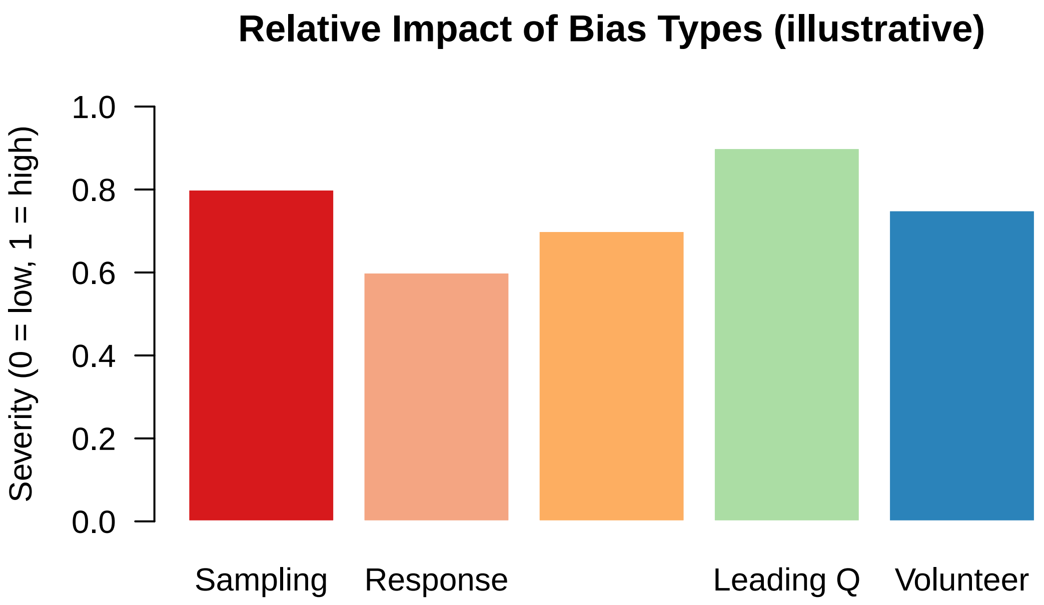

Figure 11.1.: Different types of bias and how they affect results.

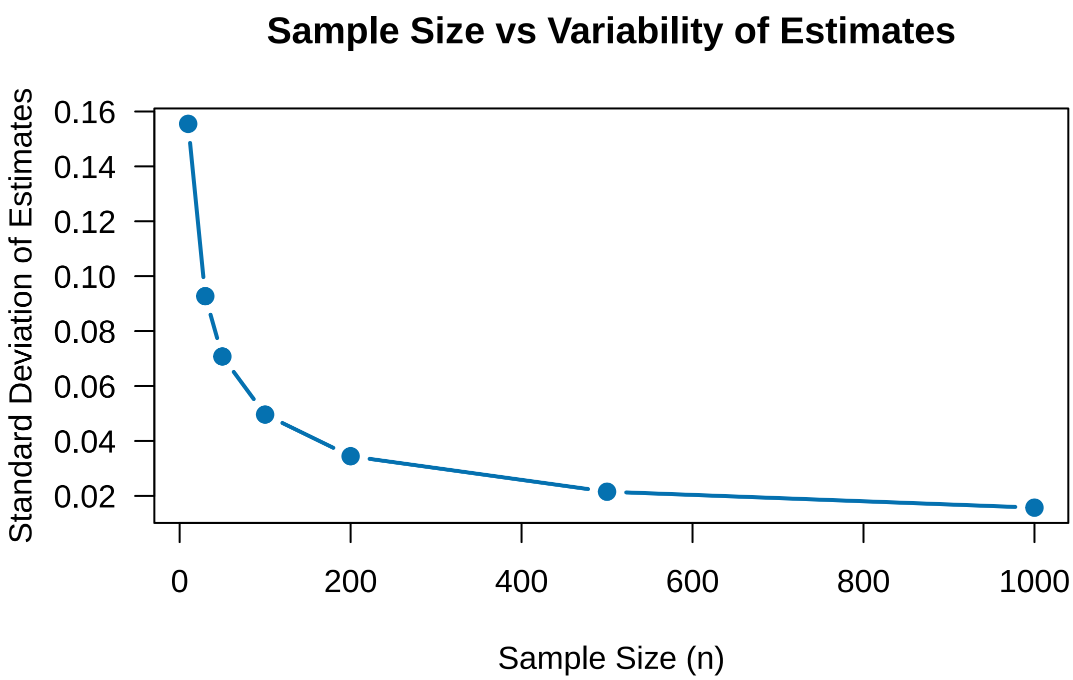

Figure 11.2.: Variability in sample estimates decreases as sample size increases.

11.4. Correlation and causation

One of the most important concepts in statistical literacy is understanding the difference between **correlation** and **causation**.

> ⚠ Correlation Causation
>
> Two variables can be **correlated** (move together) without one causing the other. There may be a **confounding variable** responsible for both.

11.4.1. Famous spurious correlations

- Ice cream sales and shark attacks both increase in summer → they are both caused by **hot weather**, not each other.
- Countries with more TVs have longer life expectancy → both are driven by **wealth**, not TVs causing people to live longer.

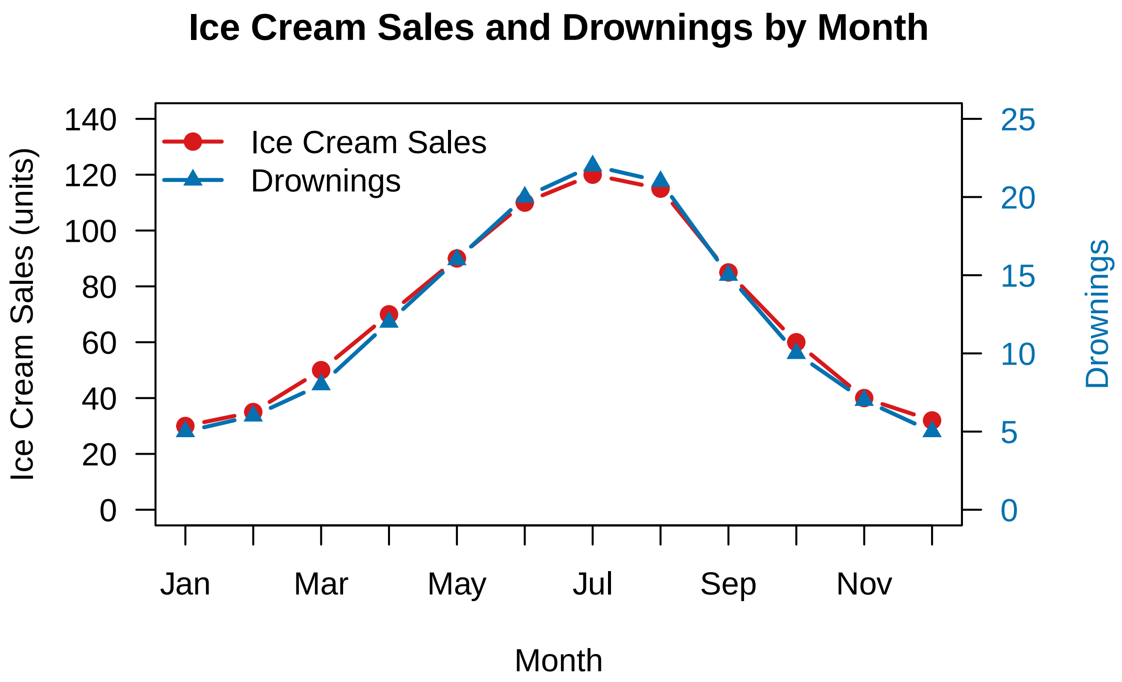

Figure 11.3.: A spurious correlation: ice cream sales and drownings both increase in summer.

11.5. Interpreting graphs critically

When reading statistical graphs, watch out for:

11.5.1. Misleading y-axis

A y-axis that doesn't start at zero can make small differences look huge.

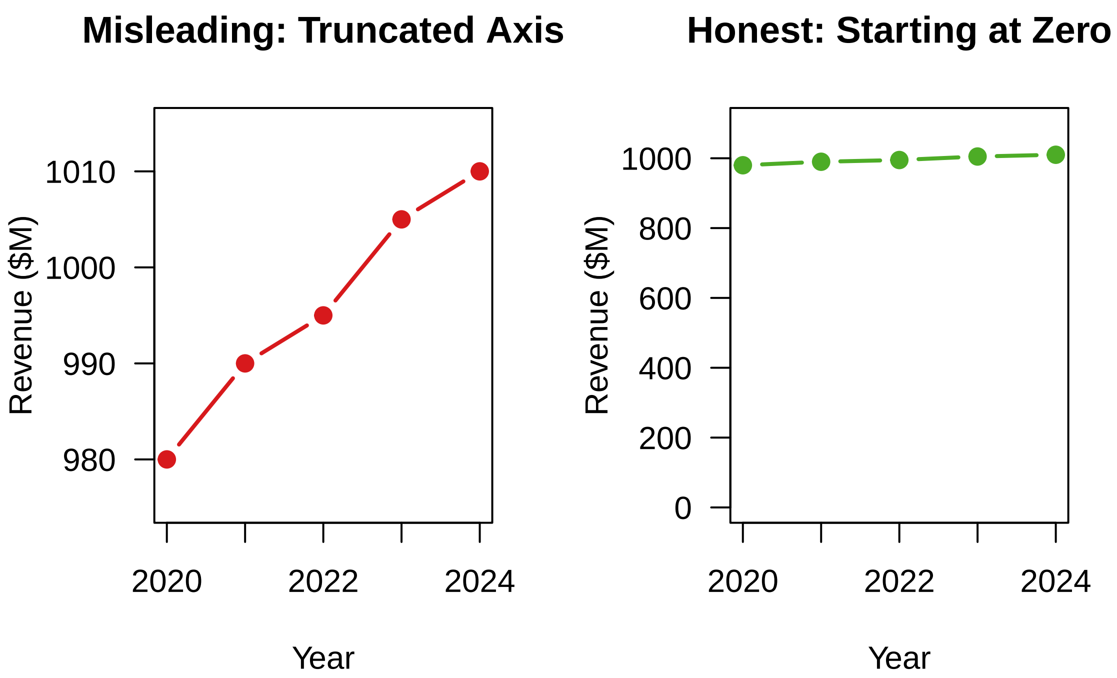

Figure 11.4.: The same data displayed with a truncated y-axis (left) vs starting at zero (right).

11.5.2. Percentage vs absolute numbers

> Be careful with percentages
>
> "Crime increased by 100%" sounds alarming — but if there were only 2 crimes last year and 4 this year, the absolute increase is tiny.
> Always ask: **100% of what?**

11.5.3. Averages without context

The **mean** can be pulled by extreme values, making it misleading if used alone. Always ask what **measure of centre** was used and whether it is appropriate.

11.6. Reading media reports — a framework

Use these questions when evaluating a statistical claim in the media:

1. **What is being measured?** (the variable and its units)
2. **Who was studied?** (the population and sample)
3. **How was data collected?** (method and potential biases)
4. **How large was the sample?**
5. **What is the source?** (government, independent researcher, company with a vested interest?)
6. **Are conclusions supported by the evidence?** Or is the claim overstated?
7. **Are comparisons fair?** Are groups being compared actually comparable?

11.7. A worked example

> **News headline**
>
> *"Study shows eating chocolate improves exam scores by 15%"*

Critical analysis:

- **What was measured?** Exam scores before and after eating chocolate.
- **Who was studied?** Not stated — red flag!
- **Sample size?** Not given — could be only 10 students.
- **Confounding variables?** Students who eat chocolate may be from wealthier families with better study resources.
- **Causation or correlation?** This is a correlational study; chocolate doesn't necessarily *cause* better scores.
- **Who funded it?** If a chocolate company funded it, there is a conflict of interest.

Conclusion: The headline is likely exaggerated or misleading. We would need much more information before accepting this claim.

11.8. Glossary

Term	Definition
Bias	Systematic error that distorts results
Confounding variable	A hidden variable that explains an apparent relationship
Correlation	A statistical relationship between two variables
Causation	One variable directly causes a change in another
Reliable	Consistent results if repeated
Valid	Measuring what it claims to measure
Population	The full group of interest
Sample	A subset of the population

11.9. Check your understanding

1. A newspaper article claims: "People who eat breakfast earn more money than those who skip it." Identify a possible confounding variable.
2. A survey found that 80% of students support a longer lunch break. The survey was conducted by standing outside the school canteen. What type of bias is this? How might it affect results?
3. A graph shows the number of electric vehicles sold in New Zealand, with the y-axis starting at 45,000 rather than 0. Sales increased from 47,000 to 51,000. On the graph, this looks like a 400% increase. Calculate the true percentage increase.
4. Explain the difference between correlation and causation using your own example.

Answers

1. Wealthier people are more likely to eat breakfast (they can afford food) and also earn more — wealth is the confounding variable.
2. Sampling bias / volunteer bias — students near the canteen are more likely to want a longer lunch break. Results will overestimate support.
3. True % increase = $(51{,}000 - 47{,}000)\ /\ 47{,}000 \times 100 \quad 8.5\%$.
4. Accept any reasonable example, e.g., "Countries with more hospitals have higher death rates — hospitals don't cause death; both are linked to population size."

12. Resource Library

This section consolidates the most valuable external resources used across this book. Each item includes a direct link and its bibliography citation key.

12.1. Assessment And Standard Documents

- Achievement Standard 91946 (NZQA PDF) (New Zealand Qualifications Authority 2024a)
- NZQA Main Site (New Zealand Qualifications Authority 2024b)

12.2. New Zealand Curriculum Resources

- The New Zealand Curriculum (Ministry of Education New Zealand 2007)
- Mathematics And Statistics Achievement Objectives (Ministry of Education New Zealand 2023)

12.3. Official Data Sources

- Stats NZ (Statistics New Zealand 2024)
- CensusAtSchool New Zealand (University of Auckland 2024)

12.4. Software And Technical Resources

- R Project (R Core Team 2024)
- Quarto (Allaire et al. 2024)
- ggplot2 Book (Wickham, Navarro, and Pedersen 2023)

12.5. Statistical Thinking And Literacy Texts

- Chance Encounters: A First Course in Data Analysis and Inference (Wild and Seber 2000)
- Towards an understanding of statistical thinking (Pfannkuch and Wild 2004)
- How to Lie with Statistics (Huff 1954)
- Naked Statistics: Stripping the Dread from the Data (Wheelan 2013)

12.6. Citation Notes

The full bibliographic records for this section are maintained in references.bib.

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Achievement Standard

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